

Fiscal Consolidations in Commodity-Exporting Countries: A Small Open Economy DSGE Perspective*

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Abstract

Fiscal consolidation in commodity-exporting economies is inherently uncertain because fiscal revenues could depend heavily on volatile international commodity prices. This paper develops a small open economy Dynamic Stochastic General Equilibrium (SOE DSGE) model to quantify how deviations in commodity revenues affect consolidation outcomes, capturing key fiscal instruments such as distortionary taxes, public spending, and external borrowing. The model is calibrated and estimated using Ecuador’s 2020–2025 IMF-supported program, an illustrative case of an oil-dependent economy pursuing fiscal adjustment under external assistance. Simulation results show that historically plausible fluctuations in oil revenues substantially alter fiscal and macroeconomic dynamics: higher-than-expected revenues improve the deficit and lower the risk premium, potentially weakening incentives to sustain consolidation, while revenue shortfalls worsen the fiscal balance, raise financing costs, and intensify the required adjustment effort. These asymmetries underscore the vulnerability of consolidation programs to exogenous shocks and the need for frameworks that explicitly incorporate revenue uncertainty. The analysis emphasizes the importance of incorporating revenue uncertainty into the design of fiscal programs and highlights the potential value of flexible targets or contingency clauses to improve their credibility and resilience. Although the calibration is specific to Ecuador, the model and findings are broadly relevant to other resource-rich economies facing similar challenges in balancing fiscal sustainability with exposure to volatile commodity markets.

Keywords: Fiscal consolidation, commodity exporter, small open economy, DSGE model

JEL classification: E62, F34, F47, Q43

*The views expressed in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or the Central Bank of Chile.

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